

Pharmacology: Drug Toxicities, Pharmacogenomics, and Drug Interactions

Practice Quiz

❖ Pharmacology bridges physiology, pathophysiology, and clinical practice.

CHALLENGE: Cover the options and try to answer the questions without looking at the choices.

Answers can be found by a SEARCH of the Notes handouts.

THE ANSWER KEY WILL BE POSTED FRIDAY NOV. 21.

PRELIMINARY RECALL QUESTIONS

How many risk factors for adverse drug reactions can you WRITE DOWN on a piece of paper or on e-paper?

Hint: You can find the answer in the PowerPoint.

NOW: Write in as many mechanisms of adverse drug effects that you think of for the following patient groups?

FILL IN

Older patients	
Pregnant patients	
Pediatric patients	
Patients with reduced organ function	
Patients on several concomitant medications	

1. A 15-year-old female prescribed oral tetracycline for acne is cautioned not to take it together with dairy products, antacids, iron, calcium, or magnesium supplements, which directly bind to the drug.

What is the mechanism of this interaction?

- A. Inhibition of absorption
- B. Inhibition of distribution
- C. Inhibition of metabolism
- D. Inhibition of excretion

2. A 67-year-old male presents to the internal medicine clinic with complaint of 8 to 10 watery stools per day for the past 7 days. He denies fever, blood in the stools, abdominal pain, and loss of appetite. His medications are a daily proton pump inhibitor for 6 months for self-diagnosed gastric reflux, daily Aleve (naproxen sodium) for self-diagnosed “arthritis pain”, and oral levofloxacin (antibiotic) for 14 days prescribed for a recurrent sinus infection, which was completed 10 days ago.

PMH: Appendectomy 3 months ago.

Lab: Stool samples are positive for *Clostridioides difficile* infection (formerly named *Clostridium difficile*).

How would this adverse drug event (ADE) be classified? (more than one answer)

- A. Dose related
- B. Duration related
- C. Drug withdrawal
- D. Failure of therapy
- E. Not related to dose
- F. Time-related, Acute
- G. Time-related, Delayed
- H. Common and predictable
- I. Rare and unpredictable

Suggestion: Highlight the clues in the stem.

Information to help you think through the question:

Microbiology: *C. difficile* is an anaerobic, gram-positive, spore-forming, toxin-producing bacillus that exists in spore and vegetative forms. It is highly transmissible by the fecal oral route and can be acquired in the hospital, where it can be cultured from hospital beds and other surfaces, as well as hands, clothing, and stethoscopes of health care workers. Either the spores or the vegetative form of the organism can be transmitted. *C. difficile* occupies the colon and is resistant to nearly all antibiotics.

Risk factors for infection:

- Antibiotics (especially broad-spectrum drugs like levofloxacin) disrupt the intestinal microbiome, allowing overgrowth of the non-susceptible microorganisms, such as *C. difficile*, which proliferates and expresses exotoxins. The toxins cause diarrhea and colitis.
 - *C. difficile*-associated diarrhea can occur days, weeks, even months, after completion of the antibiotic therapy.
 - Risk for *C. difficile* infection is increased with a longer duration of antibiotic therapy.
- Gastric acid suppression with proton pump inhibitors is associated with increased risk of *C. difficile* infection.

- NSAID use (naproxen sodium) can cause GI toxicity, which can sensitize GI epithelial cells to damage by pathogenic toxins.
- Older age (≥ 65 years of age) seems to promote increased frequency and severity of the infection.

3. A pregnant 20-year-old person (she/her/hers) with a 2-year history of regular heroin abuse is admitted to the labor and delivery ward. She gives birth to a 6 lb, 2 oz boy, gestational age 40 weeks, by vaginal delivery. There are no signs of malformations. The baby exhibits hypertonia, tremors, high-pitched crying, poor sucking reflex, yawning, and autonomic instability (fever, temperature instability).

What is the most likely cause of the newborn's signs/symptoms?

- A. Effect of Vitamin K shot given at birth
- B. Neonatal abstinence syndrome
- C. Shock of coming out of warm, dark womb into a cold, bright room
- D. Teratogenic effects of the drug

4. A 26-year-old female presents to the family practice clinic with complaint of fever, severe left ear pain, and muffled hearing in the affected ear. She reports that she had cold symptoms with a runny, stuffy nose and cough for the last week. Antibiotic history is pertinent for penicillin for the treatment of strep throat 10 years ago, which she tolerated well. NKDA (no known drug allergies). Otoscopy shows bulging, erythematous tympanic membrane and reduced mobility of the tympanic membrane. An e-prescription is submitted for amoxicillin-clavulanate 875/125 mg by mouth twice daily (po BID).



Hives (urticaria) on black skin

<https://www.healthline.com/health/skin/hives-on-black-skin>

Thirty minutes after taking the first dose, the patient develops a generalized itchy rash (urticaria) and facial swelling (angioedema).

What type of adverse drug reaction is described? Short answer

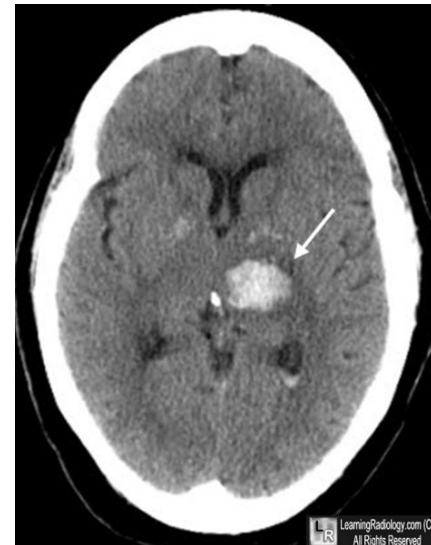
5. A 70-year-old female (pronouns she, her, hers) presents to the ED with intense, burning pain in her head, arms and legs, and signs/symptoms of a stroke. Medical history is significant for poorly controlled hypertension for ten years despite three antihypertensive medications and atrial fibrillation diagnosed 2 years ago for which she takes the anticoagulant warfarin.

Diagnostic studies:

- Findings on head CT without contrast are consistent with an acute intracerebral hemorrhage into the thalamus.
- Laboratory test shows elevated PT/INR, which is indicative of high blood levels of warfarin, which has a narrow therapeutic window.

Which of the following options describe the type of adverse drug effect? (more than one answer)

- A. Off-target effect
- B. On-target effect
- C. Intended tissue
- D. Unintended tissue
- E. Predictable effect
- F. Unpredictable effect
- G. Dose-related



Learning Radiology
<http://learningradiology.com/notes/general/notes/hemorrhagicstroke.htm>

6. A 14-year-old male is given a diagnosis of seizure disorder and evaluated for treatment with the antiseizure drug, carbamazepine. It is the preferred drug for this patient based on his type of seizure, efficacy, side effect profile, and cost. The patient's parents are immigrants from Bangalore, India. Pharmacogenomics studies indicate that this patient may be at increased risk of developing a hypersensitivity reaction characterized by potentially life-threatening cutaneous reactions.

How might the risk be minimized?

- A. Screen for HLA genetic variants.
- B. The reaction occurs rarely (up to 10% of patients) so standard dose should be prescribed.
- C. Start with low dose, titrate up, and monitor therapy.
- D. Select a different drug.

7. Carbamazepine is an inducer of CYP3A4 metabolic enzymes and can *induce its own metabolism* (autoinduction).

i) Which of the following options is the most appropriate method for optimizing carbamazepine treatment for seizure control in the individual patient?

- A. Therapeutic plasma levels
- B. Electroencephalograms
- C. Observation for seizure activity
- D. Signs and symptoms of skin rash

ii) What does “induce its own metabolism” mean in terms of drug effects over time?

8. A 38-year-old woman complains to her doctor that she has been feeling sad, hopeless, and irritable for about 6 months. She has lost interest in everything. She relates that her family and co-workers have been urging her to get help. NKDA. After a full workup, she is given a diagnosis of major depressive disorder. An appointment is made with a mental health counselor and an electronic prescription (e-prescription) is submitted for an antidepressant that blocks the reuptake transporter of a specific neurotransmitter in the central nervous system and periphery. The antidepressant described in the vignette also binds the same type of receptors in the periphery and other types of receptors in both the periphery and the CNS.

i) Which one of the following options can be inferred about the drug’s properties? (more than one answer)

- A. The drug has a high lipid:water partition coefficient.
- B. The drug has high affinity for plasma proteins.
- C. The drug has high selectivity.
- D. The drug has high potency.
- E. The drug has high potential for side effects

ii) What are the mechanisms of adverse effects?

9. A 48-year-old female presents to her PCP complaining of continuous, non-radiating muscle soreness in both thighs and calves for two days, which then worsened to crampy-type muscle pain (8 on a scale of 0-10) starting 4 hours ago. Three days ago, the patient went to the urgent care because of a “bad cough in my chest for one week but no fever”. The antibiotic clarithromycin was prescribed, and she states she has been taking it as directed. The patient’s chronic medications are lisinopril for hypertension and high-intensity simvastatin for hypercholesterolemia. Review of her medical record shows complaints on routine visits of minor intermittent muscle aches and muscle cramping since starting the statin therapy. The patient is usually healthy, active, and happy with her life.

Labs: Serum creatine kinase (CK) level is slightly elevated. Urinalysis is negative for myoglobinuria, proteinuria, and hematuria.

If the signs and symptoms are related to her drug therapy, how could this adverse event have been avoided?

- A. Check for potential drug interactions before prescribing concomitant drugs.
- B. Do not prescribe antibiotics for viral illness (“a cough without fever”).
- C. If antibiotic therapy with a CYP inhibitor is required, hold the statin until antibiotic completed.
- D. All options are correct.

Information:

CK (also called creatine phosphokinase, CPK) is an indicator of skeletal muscle cell lysis.

Simvastatin is a CYP3A4 substrate.

High-intensity = high dose

Simvastatin can cause concentration-dependent muscle toxicity.

Clarithromycin is a CYP3A4 inhibitor.

10. A 22-year-old male with schizophrenia resistant to first-line antipsychotic agents is initiated on treatment with clozapine, a highly effective drug for this indication but has a potential for causing serious adverse effects. The patient smokes two packs of cigarettes per day (10 pack-years). The clozapine dosage will be slowly titrated to the therapeutic plasma level. Because the patient is a smoker, a higher dose may be needed compared to nonsmokers. Monitoring signs and symptoms of serious toxicities including severe neutropenia, myocarditis and other cardiac anomalies, and more.

i) Which of the following metabolic enzymes is involved in this drug-smoking interaction?

- A. CYP1A2
- B. CYP2B6
- C. CYP2C19
- D. CYP2D6
- E. CYP3A4

ii) Why might this patient need a higher dose to achieve serum levels within the therapeutic range than a patient who does not smoke?

Hint: the mechanism of the smoking-drug interaction

11. The anti-hepatitis B virus (HBV) drug adefovir and the anti-HIV and anti-HBV drug tenofovir are both analogs of the nucleoside, adenosine. Here are the drug properties:

- They both are manufactured as prodrugs that facilitate their abilities to cross intestinal epithelial cell membranes by passive diffusion.
- They are both hydrolyzed to the active drug by different mechanisms.
- They are both converted by intracellular kinases to the phosphorylated form.
- They are both secreted as active drugs into the nephron via OATs.
- They both target the active site of the viral polymerase and compete for the same natural substrate.
- They are both associated with serious renal toxicity.

i) List all the potential mechanisms of this drug-drug interaction.

ii) What are the predictable clinical implications of combined therapy with adefovir and tenofovir?

11. A 24-year-old woman presents to her PCP with a “bad sunburn” after being on beach for only one hour. The affected area is painful and interrupting her sleep. She admits that she ignored the doctor’s and pharmacist’s caution to avoid exposing herself to the sun while taking doxycycline. A bright red, confluent rash on her face, neck, upper chest, back, legs, and arms in sun-exposed areas is noted on physical examination. Welts and blistering are absent. She is advised to apply cold compresses to the affected area to relieve the symptoms and take acetaminophen for the pain.



UpToDate Graphic 86479 version 8.0

What type of adverse drug reaction is described?

Please explain the mechanism.

12. A patient is given a chemotherapy regimen for the treatment of breast cancer. The drugs directly damage the tumor cells’ DNA and the DNA synthesis processes. The chemotherapy triggers the cell death pathway. The tumor cell fragments and dies without disrupting adjacent tissue.

What is the designation of the described process?

- A. Apoptosis
- B. Autophagy
- C. Immune-mediated cell death
- D. Necrosis

13. An 86-year-old man with valvular heart disease taking the anticoagulant warfarin (a vitamin K antagonist) presents to his cardiologist because of a laboratory finding of significantly elevated PT/INR of 14 (normal therapeutic range 2-3). He reports that he has not had nose bleeds or bleeding gums. The patient was hospitalized for management of intracerebral bleeding 8 months ago. Bruises are noted on bilateral upper arms. Signs and symptoms of serious bleeding are absent. Given the patient's high bleeding risk, he is given the reversal agent, oral vitamin K, and directed to stop taking the warfarin for now. He is to return to the office tomorrow for PT/INR testing.

What is the DIRECT action of vitamin K in this context?

- A. Outcompeting warfarin at vitamin K reductase
- B. Preventing serious bleeding event
- C. Supplementing dietary intake of vitamin K

Information on the drug mechanism: Warfarin is a competitive antagonist of the enzyme necessary for activating vitamin K, which is necessary for activating the synthesis of hepatic coagulation factors.

14. A 16-year-old female presenting to the women's health clinic believes she is pregnant and has been hiding it from her parents. She has not received prenatal care. She takes valproic acid for seizure disorder. Ultrasound images are consistent with 20 weeks gestation. A neural tube defect is noted.

What two terms describe this adverse drug event?

15. A 58-year-old woman with resistant hypertension is given add-on therapy with a vasodilator called hydralazine. Four weeks later she presents to the cardiologist with fever, malar rash (photo), myalgia, and arthralgia. She states she has been taking all her blood pressure medicines exactly as directed and that she is not taking any other drugs or recreational substances. There are no drug interactions with her antihypertensive medications. Suspecting a lupus-like syndrome related to elevated plasma drug concentrations due to deficient activity of the enzyme that inactivates hydralazine by acetylation, the drug is discontinued and her antihypertensive therapy is adjusted. The symptoms/signs resolve.



Picture: Journal of Clinical Pharmacy and Therapeutics
<https://onlinelibrary.wiley.com/doi/10.1111/jcpt.12506>

i) If the adverse drug effect is due to deficient activity of the drug metabolizing enzyme, which one of the following options is the term that describes this patient's pharmacogenetic phenotype?

- A. Poor metabolizer
- B. Intermediate metabolizer
- C. Extensive metabolizer
- D. Ultra-rapid metabolizer

ii) Which drug metabolizing enzyme is involved?

- A. NAT2
- B. CYP2D6
- C. G6PD
- D. VKORC1

Fill the blanks (Hint: Handout p15-16)

Metabolizer Type	Enzyme Activity	Clinical Impact	
		Standard Drug	Prodrug
Poor metabolizer PM			
Intermediate metabolizer IM			
Extensive metabolizer EM			
Ultrarapid metabolizer UM			

16. A healthy 30-year-old female presenting to the obstetrician is pleased to find out that she is 8 weeks pregnant. She asks how the doctor knows if various over-the-counter and prescription medications she might need during pregnancy are safe for her unborn baby. The doctor explains the FDA has issued guidance for the use of medications in pregnancy and breastfeeding to help patients understand the potential risks and benefits of medication use.

i) What is the name of this set of regulations on drug and biological product labels?

(Hint: PLLR)

ii) What three main categories of information are provided in this FDA guidance document?

17. A 74-year-old man with a history of stable angina is celebrating his birthday. His buddy gives him a “little blue pill” (Viagra) so he can “have fun tonight”. The man takes the Viagra (sildenafil) and collapses on the floor (from hypotension). His wife calls an ambulance. The patient’s chronic anti-angina medications are a long-acting beta blocker and a long-acting nitrate.

What is the mechanism of this drug-drug interaction that resulted in the adverse drug event?

- A. Additive
- B. Competitive
- C. Functional
- D. Mutualistic

Information:

Beta-blockers slow heart rate and reduce the force of cardiac contraction, which reduce the work of the heart and lower blood pressure.

Nitrates cause dilation of veins and arteries, which reduces the work of the heart and lower blood pressure.

PDE-5 inhibitors like Viagra indirectly sustain levels of nitric oxide in vascular smooth muscle by preventing breakdown of nitric oxide.

18. A 64-year-old woman is prescribed an oral anticoagulant called apixaban for treatment of atrial fibrillation. On questioning she reports having a glass of grapefruit juice every day for breakfast and dinner. She is cautioned that grapefruit juice can increase the plasma levels of apixaban, which could lead to serious bleeding, including bleeding in the brain. The patient agrees to switch to orange juice.

What enzymes metabolize apixaban?

19. A 54-year-old woman with a history of tophaceous gout manages hyperuricemia with probenecid, a competitive inhibitor of uric acid transport in the proximal tubule, which increases uric acid excretion. Probenecid also inhibits several isoforms in the same super family of proteins, which reduces excretion of drugs, such as penicillin, cephalosporins, ibuprofen, and many others.

From the description, you can deduce what protein is inhibited by probenecid.

What is it?

- A. CYP3A4
- B. Na^+, K^+ -ATPase
- C. Organic anion transporter
- D. Organic cation transporter
- E. P-glycoprotein

Definitions: tophaceous – rough, stony; tophaceous gout – uric acid crystals (tophi) deposit in soft tissues and form hard nodules.

20. JB is a 35-year-old man with a 10-year history of HIV infection well managed with antiretroviral therapy. He is a health care worker in a group home where he has been in close contact with a resident who has been diagnosed with meningococcal meningitis, which is caused by *Neisseria meningitidis*. One of the antibiotic options for chemoprophylaxis is a strong CYP3A4 inducer that would decrease the efficacy of his anti-HIV therapy.

What is the drug given to prevent the CNS infection?

- A. Ceftriaxone
- B. Ciprofloxacin
- C. Erythromycin
- D. Rifampin
- E. Tetracycline

21. A 38-year-old female presents to the ED with swelling, pain, redness, and warmth in her left lower extremity for three days. She describes the pain as a deep soreness (7 on a scale of 1 to 10) that increases with physical activity. She states that she tried cold compresses, elevation, ibuprofen, and acetaminophen. Nothing made it better or worse. The patient has a 10-year history of bipolar disorder stabilized on carbamazepine and lithium. She takes no other prescription drugs. She states that, as an insurance agent, she sits all day and doesn't get any exercise. She has been smoking 1-2 packs of cigarettes per day since the age of 16 (approximately 40 pack years). She admits to "a couple glasses of white wine" every evening after work. She denies use of recreational drugs. NKDA.

D-dimer and lower extremity doppler ultrasonography findings are consistent with deep vein thrombosis.

The patient is hospitalized and treated with subcutaneous low molecular weight heparin (LMWH) (an anticoagulant). In preparation for discharge, she is evaluated for treatment with an oral direct-acting anticoagulant (DOAC) metabolized to inactive metabolites by CYP3A4.



Source: Kevin J. Knopp, Lawrence B. Stack, Alan B. Storrow, R. Jason Thurman: The Atlas of Emergency Medicine, 5e Copyright © McGraw Hill. All rights reserved.

i) What is the likely effect of the concomitant administration carbamazepine and the oral anticoagulant?

- A. Decreased blood levels of carbamazepine
- B. Decreased blood levels of the oral anticoagulant
- C. Increased blood levels of carbamazepine
- D. Increased blood levels of the oral anticoagulant

ii) Which one of the following options is the most appropriate approach to anticoagulant therapy for this patient?

- A. Discontinue the carbamazepine and prescribe DOAC therapy.
- B. Continue carbamazepine and prescribe a higher dose of the DOAC.
- C. Continue carbamazepine and prescribe self-injections of LMWH.
- D. Switch the carbamazepine to a different anti-mania drug and prescribe the LMWH.

Definitions:

- Carbamazepine is an antiseizure drug also used in the management mania of bipolar disorders.
- Lithium is an anti-mania drug excreted in the urine (and sweat and saliva) as unchanged drug.
- Heparin does not undergo phase 1 or phase 2 metabolism.
- NKDA: no known drug allergies

Hint: Highlight in the vignette the clues to help you keep track of them and then identify the answers.

TEST YOUR KNOWLEDGE OF MECHANISMS AND EFFECTS

Test-taking skills:

The purpose of this exercise is to help you differentiate the information (clues) you need to identify the correct option in case vignette MCQs from the information you do not need.

In the following examples, the name of the drug, action, and therapeutic uses are given. Please identify the mechanism of the drug interaction and potential clinical effect.

YOU DO NOT NEED TO MEMORIZE THE DRUG NAMES AT THIS TIME – JUST THE MECHANISMS OF DRUG INTERACTIONS.

Drug combination	Mechanism, Effect, and Action to mitigate effect
· Omeprazole increases gastric pH for treatment of gastroesophageal reflux · Itraconazole is an antifungal agent that requires an acid pH for absorption.	
· Omeprazole is a gastric acid suppressant and strong CYP2C19 inhibitor. · Clopidogrel is an antiplatelet prodrug that is converted to its active form by CYP2C19.	
· Diazepam is a positive allosteric modulator that potentiates the effects of the neurotransmitter GABA. It is an anxiolytic, sedative, and antiseizure drug. · Alcohol (ethanol) is a recreational beverage and CNS depressant by its action as a GABA-A receptor positive allosteric modulator.	
· Cholestyramine is a resin that is not absorbed from the GI tract and forms a complex with bile acids and many drugs. · Many drugs	

· Naproxen (Aleve) is a nonsteroidal anti-inflammatory drug (NSAID) for pain relief. It reduces platelet function. · Apixaban is a direct oral anticoagulant (DOAC) for treatment and prevention of thromboemboli (clots).	
· Nirmatrelvir-ritonavir (Paxlovid) is an antiviral agent for Covid treatment. Ritonavir is a powerful CYP3A4 inhibitor. · Many drugs are metabolized by CYP3A4.	
· Methotrexate is for the treatment of rheumatoid arthritis (and other autoimmune diseases). Serious AEs include hepatotoxicity and bone marrow toxicity. · Leflunomide is for rheumatoid arthritis by a different mechanism. It is toxic to the liver and bone marrow.	
· Azithromycin is an antibiotic that inhibits P-glycoprotein efflux pump. · Digoxin increases the force of myocardial contraction (cardiotonic). It is a substrate of P-glycoprotein and has a narrow therapeutic window (small TI).	
· Propranolol is a nonselective beta blocker for the treatment of hypertension, angina, and certain arrhythmias. It can cause bronchospasm in susceptible patients. · People with asthma have airway hyperresponsiveness – an exaggerated contractile response of the airways to a variety of stimuli.	
· Penicillin is secreted as unchanged drug into the nephron via OAT. Penicillin has a very short duration of action. · Probenecid inhibits OAT	